

Chartered Institute of Technology

Model Paper (B.Tech- IInd Sem)

Unit -2

Part -A(Very Short Ans Type Question)

1. Find directional derivative of $\phi = x^2 - 2y^2 + 4z$ at $(1,1,-1)$ in the direction of $2\hat{i} + \hat{j} - \hat{k}$.
2. Evaluate $\frac{d}{dt} \left(\vec{r} \times \frac{d\vec{r}}{dt} \right)$
3. Discuss the curl point of function.
4. If $\vec{A} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$ find $\text{div}\vec{A}$ at point $(1,-1,1)$.

Part -B(Short Ans Type Question)

1. Prove that $\nabla^2 r^n = n(n+1)r^{n-2}$ OR $\nabla \cdot (\nabla r^n) = n(n+1)r^{n-2}$
2. Find the value of a, b, and c, if $\vec{A}$ is irrotational, where $\vec{A} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$
3. Find directional derivative of $\phi = x^2 - y^2 + 2z^2$ at $(1,2,3)$ in the direction of the line PQ, where Q is the point $(5,0,4)$.
4. A particle moving on the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the component of velocity and component of acceleration at $t=1$ in the direction of $\hat{i} - 3\hat{j} + 2\hat{k}$
5. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then find $\text{div}(r^n\vec{r})$ and $\text{curl}(r^n\vec{r})$.

Part -C (Long Ans Type Question)

1. A vector field $\vec{F} = \sin y \hat{i} + x(1 + \cos y) \hat{j}$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ over the circular path

given by $x^2 + y^2 = a^2$, $z = 0$.

2. Find the work done in the moving path in the force field

$\vec{F} = 3x^2 \hat{i} + (2xz - y) \hat{j} + z \hat{k}$ along

(i) the straight line from (0, 0, 0) to (2, 1, 3)

(ii) the curve defined by $x^2 = 4y$, $3x^2 = 8z$ from $x = 0$ to $x = 2$

3. Evaluate (i) $\iint_S \text{Curl} \vec{F} \cdot \hat{n} dS$ (ii) $\iint_S \phi \hat{n} dS$, where $\vec{F} = (x + 2y) \hat{i} - 3z \hat{j} + x \hat{k}$ and

$\phi = 4x + 3y - 2z$ and S is the surface $2x + y + 2z = 6$ bounded by $x = 0$, $x = 1$, $y = 0$, $y = 2$.